

SUKKUR IBA UNIVERSITY

Department of Computer Systems Engineering



Final Year Project Proposal

Submission Date:_____

Project Title:	AI-Powered Centralized Fashion System Using Web Scrapping and Diffusion Model Based Virtual Try-on		
Main Subject Area/s:	☐ Medical ☐ Agriculture ☐ Power Sector ☐ IOT ☐ Robotics ☒ Artificial Intelligence ☒ Machine Learning ☒ Software development ☒ Web Development Others_____		
Nature of Project:	☐ Research, Development, and Engineering (RD&E) leading to production capability ☒ Application-oriented Research, Design & Development (R, D&D) having production potential ☐ Basic R&D		
Mapped SDG:	1. SDG 8: Decent Work and Economic Growth. 2. SDG 9: Industry, Innovation, and Infrastructure. 3. SDG 12: Responsible Consumption and Production.		
Students:	Name		CMS ID
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Supervisor:	Engr. Umair Ayaz Kamangar	Co-Supervisor:	Engr. Dr. Abdul Sattar Chan
Industrial Supervisor:		Organization:	
Is this project a part of vertical integrated project?	Yes ☐ No ☒		
If yes, please give the title of the previous project:			
Funding availability for the project:		Yes ☐ No ☒	

If yes, please give details from which funding agency:

Industrial collaboration for this project:

Yes ☐

No ☒

If yes, please give details:

Name of collaborator:

Designation:

Company/Industry:

Is this project targeted at solving any industry- or society-related problem in particular?

Yes ☒

No ☐

If yes, please give details:

- Enhances online shopping experiences through Virtual Try-On
- Reduces need for physical samples, minimizing waste
- Offers innovative solutions that could lead to economic growth within the tech-driven fashion sector.

Project Overview

In this era of the internet, online shopping for clothing has emerged as an important feature of present-day consumer behavior. Thanks to the growth of e-commerce platforms, one can easily shop for clothes without leaving their homes, removing the need to go to physical shops. This change not only saves people's time but also provides a wider selection of options, allowing consumers to discover modern fashion trends. Moreover, integration of features like virtual try-ons really enhances the overall shopping experience.

Virtual clothing try-on has proved to be a vital feature in online shopping by allowing customers to visualize how different clothing would look and fit on them without physical interaction. Our centralized shopping system is being proposed to make it more convenient for the user to compare products from different websites and also try those products on themselves. The user will be able to see the required clothes from multiple local brand stores on a single website. We intend to achieve this by using web scraping techniques.

For the web scraping phase, we intend to use ParseHub. It's a web scraping tool that enables one to collect data and information like text, images, links etc from any website no matter what structure. Later when data has been retrieved, the user can select a product to buy or even virtually try it. We intend to accomplish try-on feature by using a model named IDM VTON which stands for improved diffusion model. Moreover, the centralized clothing shopping system makes sure that users have an effortless experience when shopping by strengthening users through latest product updates, product sales and price from local online stores.

Aims and Objectives of Project

The primary purpose of the project is to develop a system that centralizes clothing shopping experience with products from local online stores, featuring a virtual try-on tool to provide users with the option to virtually try the product based on web scraping techniques, NLP processing, and diffusion model.

Objectives:

- Identify the user's preference by filtering keywords from user input using natural language processing techniques.
- Scrap and retrieve product data based on user's preferences from local online stores/brands using ParseHub tool.

- Create a system to integrate the specified product with diffusion model to provide a virtual try-on option.

Literature Review

An important tool of data science, Web scraping otherwise called web crawling or web data mining, is a method to collect/retrieve data and information from the world wide web and record it in any database or somewhere for review and future processing[1]. In addition to this, Web scraping has also proved to be important tool to retrieve data from multiple websites or online sources. The web scraping tool Parsehub is quite easy to use and offers a wide range of options or filters for our relevance [2].

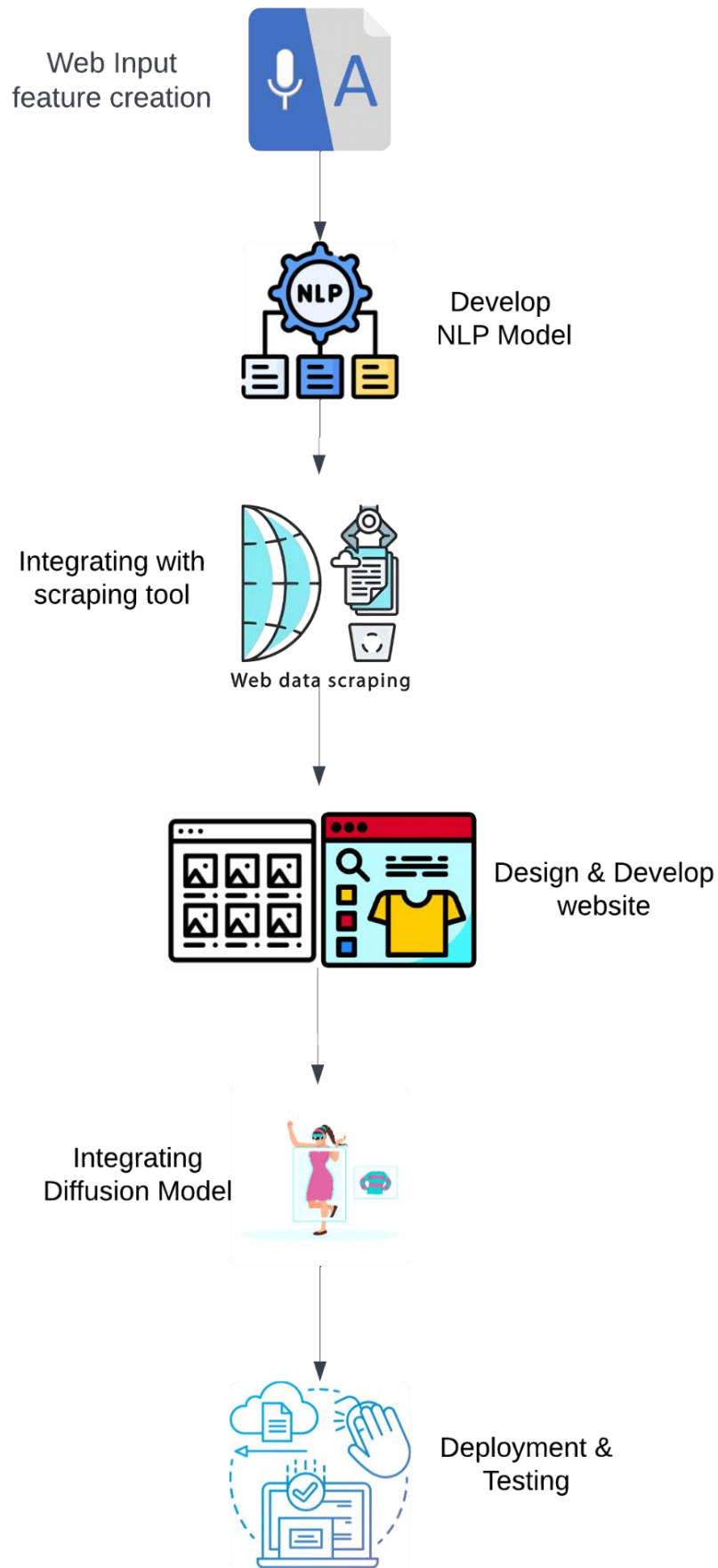
Image-based virtual try-on (VTON) is a crucial domain of computer vision where the aim is to display a visual representation of a person in a garment using given images [3, 4].

The main leading techniques for image based VTON utilize Generative Adversarial Network (GANs) [5, 6,]. However, these techniques have difficulty in generating detailed images and mostly do not adapt to a range of human visuals or photos[7, 8, 9].

On the other hand, recent diffusion models[10, 11] demonstrated improved performance in generating detailed and realistic images [12, 13, 14, 15] compared to GANs, showing a positive sign to overcome mentioned challenges. Recent diffusion-based VTON techniques[16, 2] employ the advanced generative framework of the pre-trained text-to-image (T2I) diffusion model to improve the realism of try-on visuals. However, those methods struggle to preserve the clothing characteristics, e.g design, color, pattern, shapes or prints. To solve this problem, our system introduces a model for Virtual Try-On (IDM-VTON), which stands for Improved Diffusion Model Virtual Try-on, which notably boosts the reliability of the garment image while producing genuine virtual try-on images. Our method outperforms other methods, and refining the model significantly improves preserving the details of the garment while producing realistic and authentic try-on images.

Moreover, our centralized platform ensures a seamless shopping experience by providing real-time updates on availability from online stores. This is not only time-saving but also enables users to make more informed purchasing decisions, making the virtual try-out process much easier.

Methodology for Implementation of Project



1. Web feature creation:

Web feature creation to get voice/text input from users regarding their preferences and choices like type of wear (t-shirt, hoodie, jacket, saree, wedding dress, etc.), colors, price, etc.

Develop NLP Model:

Develop a model using NLP to extract keywords related to user fashion preferences from user input, like t-shirts, sarees, or price tags like under 5000 pkr.

2. Integrating site with Scraping Tool:

Integrate the site with a web scraping tool called Parsehub in order to collect product details like product image, price, buy link, etc. from local online brands.

3. Design & Develop website:

Design the UI and develop the site with the necessary backend program to display listed products and their details to the user.

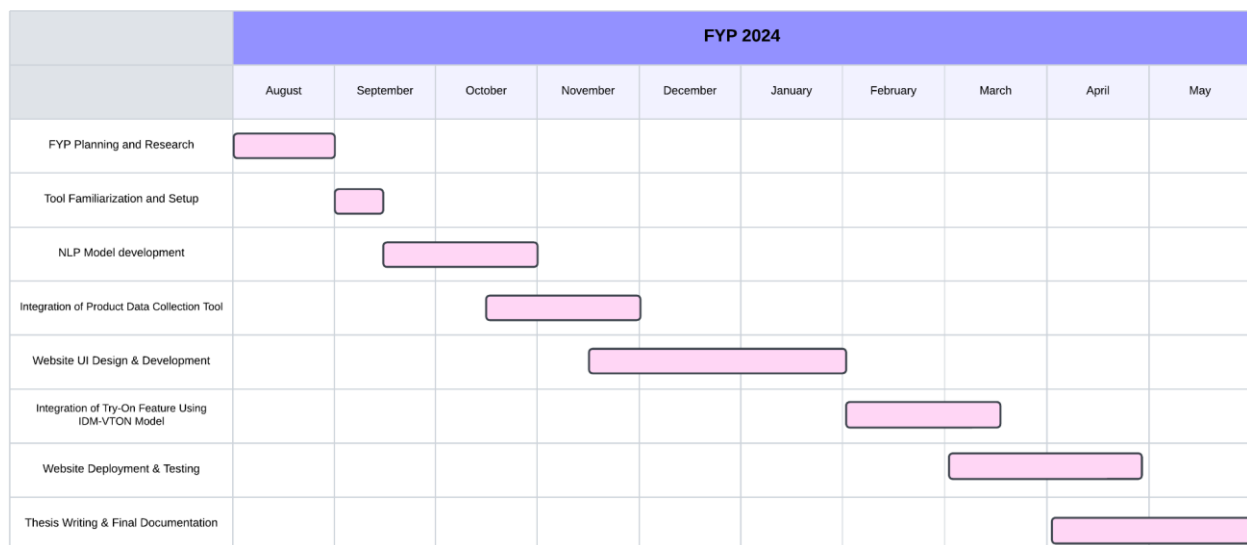
4. Integrating Diffusion Model:

Integrate the diffusion model IDM-VTON (improved diffusion model virtual try-on) with the website to offer users a virtual try-on option of the product they are interested in.

5. Deployment & Testing:

Deploy and test the site to ensure full functionality of each feature and user experience.

Project Plan/Gantt Chart



Budget Description

Write about the budget required for the proposed project here. Do mention how the equipment/experimental setup required will be arranged. E.g., an experimental setup is present in a university or some other organization's setup will be used, or funding is required for the execution of the project.

Sr. No	Item/Product	Details of Product	Source of Purchase	Quantity	Price	Total Amount
1	Web scraping API	Use API functionality of parsehub tool for products retrieval	https://www.parsehub.com/pricing	1	14000	
2	Domain register	Buy website name for online access	https://www.godaddy.com/en-ph/domainsearch/find?domainToCheck	1	15000	
3	Web hosting	To host the website application on internet	https://www.godaddy.com/en-ph/offers/hosting	1	12000	
4	Cloud storage	For storing product data on cloud	https://aws.amazon.com/s3/pricing/	1	11000	
5	Documentation	Manuals,thesis and other document printing	Print shop	1	13000	Rs 65000

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Signature of Students:

1. Tushar (Group Lead)

Signature: _____

2. Usama

Signature: _____

3. Hammad Ul Hassan

Signature: _____

Supervisor's Comments

The objectives are fine but broadly defined, so narrowing the focus might help with implementation. The proposed timeline seems to be realistic so it's fine. The methodology section is also fine but needs more detail. Overall, the proposal is good and recommended for submission as a final year project.

Name: Engr. Umair Ayaz Kamangar

Designation: Lecturer (BPS-18)

Signature: _____

FYP Evaluator Remarks

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Project Evaluator:

Name: _____

Designation: _____

Signature: _____

Final Year Project Coordinator